91. Identify a molecule which does not exist.

(1) $He_2$ (2) $Li_2$

(3) $C_2$ (4) $O_2$

**Answer (1)**

**Sol.** For $He_2$ molecule

Electronic configuration is $\sigma 1s^2$, $\sigma^* 1s^2$

$$\text{so bond order} = \frac{1}{2}[N_b - N_a]$$

$$= \frac{1}{2}[2 - 2]$$

$$= 0$$

Since, bond order is zero, so $He_2$ molecule does not exist.

92. Find out the solubility of $Ni(OH)_2$ in 0.1 M NaOH. Given that the ionic product of $Ni(OH)_2$ is $2 \times 10^{-15}$

(1) $2 \times 10^{-13}$ M (2) $2 \times 10^{-8}$ M

(3) $1 \times 10^{-13}$ M (4) $1 \times 10^{8}$ M

**Answer (1)**

**Sol.** $\underset{s}{Ni(OH)_2} \rightleftharpoons \underset{s}{Ni^{2+}} + \underset{2s}{2\overset{\ominus}{O}H}$

$\underset{0.1}{NaOH} \longrightarrow \underset{0.1}{Na^+} + \underset{0.1}{\overset{\ominus}{O}H}$

Total $[\overset{\ominus}{O}H] = 2s + 0.1 \approx 0.1$

Ionic product $= [Ni^{2+}][\overset{\ominus}{O}H]^2$

$2 \times 10^{-15} = s(0.1)^2$

$s = 2 \times 10^{-13}$

Solubility of $Ni(OH)_2 = 2 \times 10^{-13}$ M

93. Identify the correct statements from the following :

(a) $CO_2(g)$ is used as refrigerant for ice-cream and frozen food.

(b) The structure of $C_{60}$ contains twelve six carbon rings and twenty five carbon rings.

(c) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.

(d) CO is colorless and odourless gas.

(1) (a), (b) and (c) only

(2) (a) and (c) only

(3) (b) and (c) only

(4) (c) and (d) only

**Answer (4)**

**Sol.**
- Dry ice, $CO_2(s)$, is used as refrigerant
- $C_{60}$ contains 20 six membered rings, 12 five membered rings

94. Hydrolysis of sucrose is given by the following reaction.

$$\text{Sucrose} + H_2O \rightleftharpoons \text{Glucose} + \text{Fructose}$$

If the equilibrium constant ($K_C$) is $2 \times 10^{13}$ at 300 K, the value of $\Delta_r G^{\ominus}$ at the same temperature will be :

(1) $-8.314\ J\ mol^{-1}K^{-1} \times 300\ K \times \ln(2 \times 10^{13})$

(2) $8.314\ J\ mol^{-1}K^{-1} \times 300\ K \times \ln(2 \times 10^{13})$

(3) $8.314\ J\ mol^{-1}K^{-1} \times 300\ K \times \ln(3 \times 10^{13})$

(4) $-8.314\ J\ mol^{-1}K^{-1} \times 300\ K \times \ln(4 \times 10^{13})$

**Answer (1)**

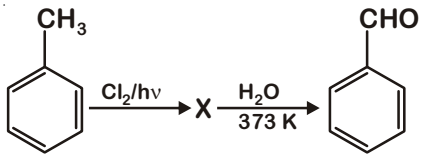
**Sol.** $\Delta G = \Delta G^\circ + RT \ln Q$

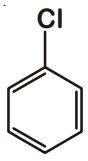
At equilibrium $\Delta G = 0$, $Q = K_{eq}$

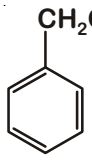
So $\Delta_r G^\circ = -RT \ln K_{eq}$

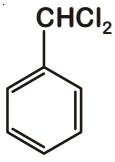
$\Delta_r G^\circ = -8.314\ J\ mol^{-1}\ K^{-1} \times 300\ K \times \ln(2 \times 10^{13})$

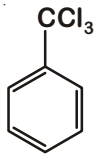
95. Identify compound X in the following sequence of reactions




(1) 

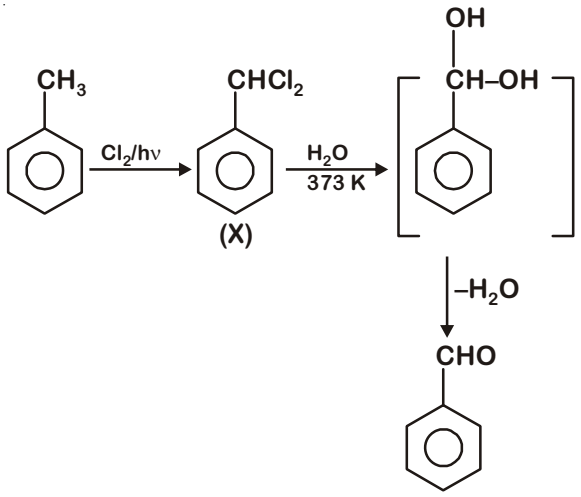

(2) 


(3) 


(4) 


**Answer (3)**

**Sol.**




96. Identify the incorrect match.

| Name | IUPAC Official Name |
|---|---|
| (a) Unnilunium | (i) Mendelevium |
| (b) Unniltrium | (ii) Lawrencium |
| (c) Unnilhexium | (iii) Seaborgium |
| (d) Ununnnium | (iv) Darmstadtium |

(1) (a), (i) (2) (b), (ii)

(3) (c), (iii) (4) (d), (iv)

**Answer (4)**

**Sol.** Ununnium

Atomic number = 111

IUPAC official name : Roentgenium

97. An element has a body centered cubic (bcc) structure with a cell edge of 288 pm. The atomic radius is

(1) $\frac{\sqrt{3}}{4}\times 288$ pm (2) $\frac{\sqrt{2}}{4}\times 288$ pm

(3) $\frac{4}{\sqrt{3}}\times 288$ pm (4) $\frac{4}{\sqrt{2}}\times 288$ pm

**Answer (1)**

**Sol.** For BCC,

$\sqrt{3}a = 4r$ $a =$ edge length

$r =$ radius

$r = \frac{\sqrt{3}a}{4}$

Given, a = 288 pm

$r = \frac{\sqrt{3}}{4}\times 288$

98. Which of the following set of molecules will have zero dipole moment?

(1) Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene

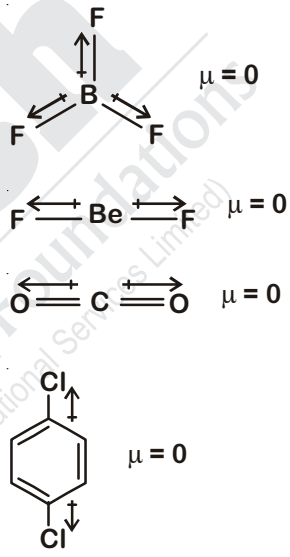
(2) Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene

(3) Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene

(4) Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene

**Answer (4)**

**Sol.**




99. On electrolysis of dil. sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be

(1) Hydrogen gas (2) Oxygen gas

(3) $H_2S$ gas (4) $SO_2$ gas

**Answer (2)**

**Sol.** During the electrolysis of dil. sulphuric acid using Pt electrodes following reaction will take place.

At cathode :

$$4H^+(aq) + 4e^- \longrightarrow 2H_2(g)$$

At anode :

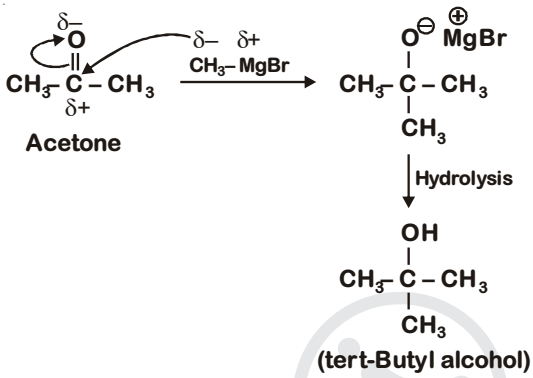
$$2H_2O(l) \longrightarrow O_2(g) + 4H^+(aq) + 4e^-$$

100. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

(1) Isopropyl alcohol (2) Sec. butyl alcohol
(3) Tert. butyl alcohol (4) Isobutyl alcohol

**Answer (3)**

**Sol.**

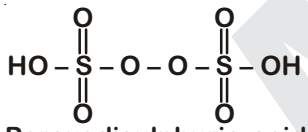



101. Which of the following oxoacid of sulphur has – O – O – linkage?

(1) $H_2SO_3$, sulphurous acid
(2) $H_2SO_4$, sulphuric acid
(3) $H_2S_2O_8$, peroxodisulphuric acid
(4) $H_2S_2O_7$, pyrosulphuric acid

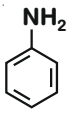
**Answer (3)**

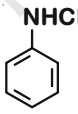
**Sol.**

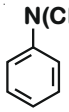


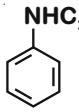

Peroxodisulphuric acid

102. Which of the following amine will give the carbyle test?

(1) 


(2) 


(3) 


(4) 


**Answer (1)**

**Sol.** Aliphatic and aromatic primary es give carbyle reaction.

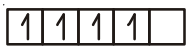
103. The calculated spin only magnetic moment of $Cr^{2+}$ ion is

(1) 3.87 BM (2) 4.90 BM
(3) 5.92 BM (4) 2.84 BM

**Answer (2)**

**Sol.** Electronic configuration of Cr – [Ar] $3d^5\ 4s^1$

Electronic configuration of $Cr^{2+}$ – [Ar] $3d^4$




Number of unpaired $e^-$ = 4

Spin only magnetic moment = $\sqrt{n(n+2)}$

n = number of unpaired $e^-$

Spin only magnetic moment = $\sqrt{4(4+2)}$

$= \sqrt{24}$ BM

= 4.9 BM

104. The correct option for free expansion of an ideal gas under adiabatic condition is

(1) $q = 0, \Delta T = 0$ and $w = 0$
(2) $q = 0, \Delta T < 0$ and $w > 0$
(3) $q < 0, \Delta T = 0$ and $w = 0$
(4) $q > 0, \Delta T > 0$ and $w > 0$

**Answer (1)**

**Sol.** Free expansion $\Rightarrow P_{ex} = 0$

$\therefore$ $w = -P_{ex}\Delta V = 0$

$\because$ Adiabatic process $\Rightarrow q = 0$

also, $\Delta U = q + w$ [ first law of thermodynamics]

$\therefore$ $\Delta U = 0$

$\because$ Internal energy of an ideal gas is a function of temperature

$\therefore$ If internal energy remains constant

$\therefore$ $\Delta T = 0$

105. The freezing point depression constant ($K_f$) of benzene is 5.12 K kg $mol^{-1}$. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places) :

(1) 0.20 K
(2) 0.80 K
(3) 0.40 K
(4) 0.60 K

**Answer (3)**

**Sol.** $\Delta T_f = k_f m$

$= 5.12\ (K.kg\ mol^{-1}) \times 0.078\ (mol\ kg^{-1})$

= 0.399 K

$\approx$ 0.40 K

106. The number of Faradays(F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass of Ca = 40 g $mol^{-1}$) is

(1) 1 (2) 2

(3) 3 (4) 4

**Answer (1)**

**Sol.** 1 equivalent of any substance is deposited by 1 F of charge.

We have, 20 g calcium

$$\text{Number of equivalents} = \frac{\text{Given mass}}{\text{Equivalent mass}}$$

$$= \frac{20}{20} = 1$$

$$\text{Equivalent mass of Ca} = \frac{40}{2} = 20$$

So, 1 faraday of charge is required.

107. Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as

(1) Aldol condensation

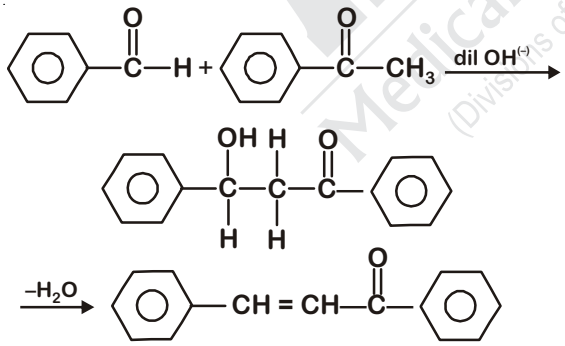
(2) Cannizzaro's reaction

(3) Cross Cannizzaro's reaction

(4) Cross Aldol condensation

**Answer (4)**

**Sol.**




In the presence of dil.$OH^{(-)}$, benzaldehyde and acetophenone will react to undergo cross-aldol condensation.

108. Paper chromatography is an example of

(1) Adsorption chromatography

(2) Partition chromatography

(3) Thin layer chromatography

(4) Column chromatography

**Answer (2)**

**Sol.** Paper chromatography is a type of partition chromatography in which a special quality paper known as chromatography paper is used.

109. An increase in the concentration of the reactants of a reaction leads to change in

(1) activation energy

(2) heat of reaction

(3) threshold energy

(4) collision frequency

**Answer (2)**

**Sol.** Heat of reaction is an extensive property. Hence, on change of amount/concentration of reactants heat of reaction changes.

110. A mixture of $N_2$ and Ar gases in a cylinder contains 7 g of $N_2$ and 8 g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of $N_2$ is :

[Use atomic masses (in g $mol^{-1}$) : N = 14, Ar = 40]

(1) 9 bar (2) 12 bar

(3) 15 bar (4) 18 bar

**Answer (3)**

**Sol.** $n_{N_2} = \frac{7}{28} = \frac{1}{4} = 0.25$

$$n_{Ar} = \frac{8}{40} = \frac{1}{5} = 0.20$$

Now, Applying Dalton's law of partial pressure, $p_{N_2} = \left(\chi_{N_2}\right) P_{Total}$

$$= \frac{0.25}{0.45} \times 27 \text{ bar}$$

$$= \frac{5}{9} \times 27 = 15 \text{ bar}$$

111. Identify the correct statement from the following :

(1) Wrought iron is impure iron with 4% carbon.

(2) Blister copper has blistered appearance due to evolution of $CO_2$.

(3) Vapour phase refining is carried out for Nickel by Van Arkel method.

(4) Pig iron can be moulded into a variety of shapes.

**Answer (4)**

**Sol.** The iron obtained from blast furnace contains about 4% carbon and many impurities like S, P, Si, Mn in smaller amount. This is known as pig iron and cast into variety of shapes.

112. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of ?

(1) – I effect of – $CH_3$ groups

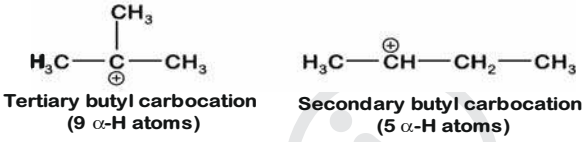
(2) + R effect of – $CH_3$ groups

(3) – R effect of – $CH_3$ groups

(4) Hyperconjugation

**Answer (4)**

**Sol.**



Tertiary butyl carbocation (9 $\alpha$-H atoms)

Secondary butyl carbocation (5 $\alpha$-H atoms)

More the number of $\alpha$-H atoms, more will be the hyperconjugation effect hence more will be the stability of carbocation.

113. Which of the following is a cationic detergent?

(1) Sodium lauryl sulphate

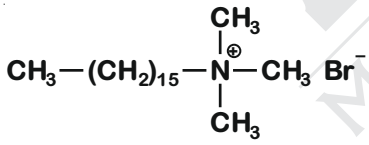
(2) Sodium stearate

(3) Cetyltrimethyl ammonium bromide

(4) Sodium dodecylbenzene sulphonate

**Answer (3)**

**Sol.**



Cetyltrimethyl ammonium bromide

114. Elimination reaction of 2-Bromo-pentane to form pent-2-ene is

(a) $\beta$-Elimination reaction

(b) Follows Zaitsev rule

(c) Dehydrohalogenation reaction

(d) Dehydration reaction

(1) (a), (b), (c)

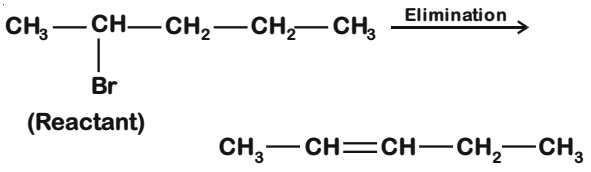
(2) (a), (c), (d)

(3) (b), (c), (d)

(4) (a), (b), (d)

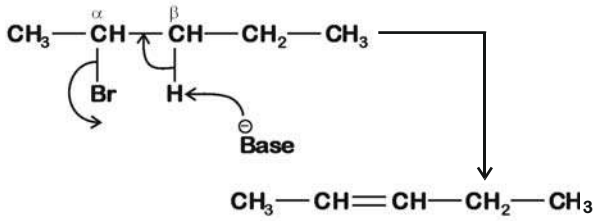
**Answer (1)**

**Sol.**



(Reactant)

Mechanism:



Since $\beta$-hydrogen is abstracted it is $\beta$-elimination.

Since more substituted alkene is formed, it follows zaitsev's rule.

Since 'H' and 'Br' are removed, it is dehydro-halogenation.

115. The mixture which shows positive deviation from Raoult's law is

(1) Ethanol + Acetone

(2) Benzene + Toluene

(3) Acetone + Chloroform

(4) Chloroethane + Bromoethane

**Answer (1)**

**Sol.** Pure ethanol molecules are hydrogen bonded. On adding acetone, its molecules get in between the ethanol molecules and break some of the hydrogen bonds between them. This weakens the intermolecular attractive interactions and the solution shows positive deviation from Raoult's law.

116. Which of the following is the correct order of increasing field strength of ligands to form coordination compounds?

(1) $SCN^- < F^- < C_2O_4^{2-} < CN^-$

(2) $SCN^- < F^- < CN^- < C_2O_4^{2-}$

(3) $F^- < SCN^- < C_2O_4^{2-} < CN^-$

(4) $CN^- < C_2O_4^{2-} < SCN^- < F^-$

**Answer (1)**

**Sol.** Spectrochemical series (as given in NCERT) :

$I^- < Br^- < SCN^{\ominus} < Cl^- < S^{2-} < F^- < \overline{O}H < C_2O_4^{2-} < H_2O < NCS^{\ominus} < EDTA^{4-} < NH_3 < en < \overset{\ominus}{C}N < CO$

**117.** Which of the following is a basic o acid ?

(1) Serine (2) Alanine

(3) Tyrosine (4) Lysine

**Answer (4)**

**Sol.** $H_2N-CH_2-CH_2-CH_2-CH_2-\underset{\underset{NH_2}{|}}{CH}-COOH$

(Structure of Lysine)

Lysine is a basic o acid.

**118.** HCl was passed through a solution of $CaCl_2$, $MgCl_2$ and NaCl. compound(s) crystallise(s)?

(1) Both $MgCl_2$ and $CaCl_2$

(2) Only NaCl

(3) Only $MgCl_2$

(4) NaCl, $MgCl_2$ and $CaCl_2$

**Answer (2)**

**Sol.** Since $CaCl_2$ and $MgCl_2$ are more soluble than NaCl, on passing HCl(g) through a solution containing $CaCl_2$, $MgCl_2$ and NaCl then NaCl crystallizes out.

**119.** Which of the following is a natural polymer?

(1) cis-1, 4-polyisoprene

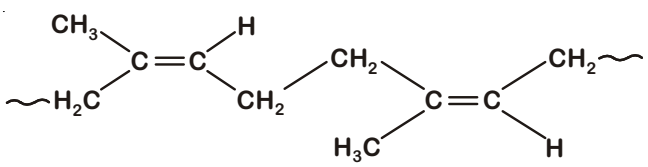
(2) poly (Butadiene-styrene)

(3) polybutadiene

(4) poly (Butadiene-acrylonitrile)

**Answer (1)**

**Sol.** Naturally occuring polymer, natural rubber is cis-1, 4– polyisoprene



**120.** Which of the following is **not** correct about carbon monoxide ?

(1) It forms carboxyhaemoglobin

(2) It reduces oxygen carrying ability of blood.

(3) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.

(4) It is produced due to incomplete combustion.

**Answer (3)**

**Sol.** The carboxyhaemoglobin is about 300 times more stable than oxyhaemoglobin.

**121.** Sucrose on hydrolysis gives

(1) β-D-Glucose + α-D-Fructose

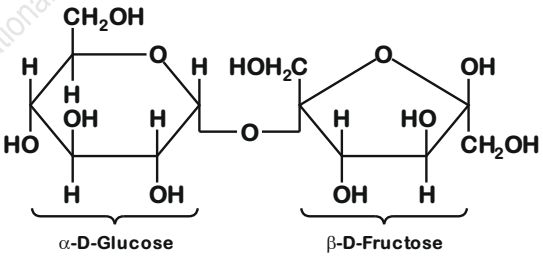
(2) α-D-Glucose + β-D-Glucose

(3) α-D-Glucose + β-D-Fructose

(4) α-D-Fructose + β-D-Fructose

**Answer (3)**

**Sol.** Sucrose $\xrightarrow{\text{Hydrolysis}}$ α-D-Glucose + β-D-Fructose




**122.** The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.

(1) Iron

(2) Copper

(3) Calcium

(4) Potassium

**Answer (4)**

**Sol.** Potassium (K) activates many enzymes participate in oxidation of glucose to produce ATP and helps in the transmission of nerve signal along with Na.

123. Which one of the followings has maximum number of atoms ?

(1) 1 g of Ag(s) [Atomic mass of Ag = 108]

(2) 1 g of Mg(s) [Atomic mass of Mg = 24]

(3) 1 g of $O_2(g)$ [Atomic mass of O = 16]

(4) 1 g of Li(s) [Atomic mass of Li = 7]

**Answer (4)**

**Sol.**

- Number of Mg atoms = $\frac{1}{24} \times N_A$
- Number of O atoms = $\frac{1}{32} \times 2 \times N_A$
- Number of Li atoms = $\frac{1}{7} \times N_A$
- Number of Ag atoms = $\frac{1}{108} \times N_A$

124. The number of protons, neutrons and electrons in $^{175}_{71}Lu$, respectively, are

(1) 71, 104 and 71 (2) 104, 71 and 71

(3) 71, 71 and 104 (4) 175, 104 and 71

**Answer (1)**

**Sol.** $^{175}_{71}Lu$

No. of Protons= 71 = No. of Electrons

No. of Neutrons = Mass no. – No. of Protons

= 175 – 71

= 104

125. What is the change in oxidation number of carbon in the following reaction?

$$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(l) + 4HCl(g)$$

(1) + 4 to + 4

(2) 0 to + 4

(3) – 4 to + 4

(4) 0 to – 4

**Answer (3)**

**Sol.** $CH_4 \Rightarrow x + 4 \times 1 = 0 \Rightarrow x = -4$

$CCl_4 \Rightarrow x + 4 \times (-1) = 0 \Rightarrow x = +4$

$$\overset{-4}{CH_4}(g) + 4Cl_2(g) \rightarrow \overset{+4}{CCl_4}(l) + 4HCl(g)$$

Change in oxidation state of carbon is from –4 to +4

126. Identify the **incorrect** statement.

(1) $Cr^{2+}$ ($d^4$) is a stronger reducing agent than $Fe^{2+}$ ($d^6$) in water.

(2) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.

(3) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.

(4) The oxidation states of chromium in $CrO_4^{2-}$ and $Cr_2O_7^{2-}$ are not the same.

**Answer (4)**

**Sol.** Oxidation state of Cr in $CrO_4^{2-}$ and $Cr_2O_7^{2-}$ is +6

127. For the reaction, $2Cl(g) \longrightarrow Cl_2(g)$, the correct option is :

(1) $\Delta_r H > 0$ and $\Delta_r S > 0$

(2) $\Delta_r H > 0$ and $\Delta_r S < 0$

(3) $\Delta_r H < 0$ and $\Delta_r S > 0$

(4) $\Delta_r H < 0$ and $\Delta_r S < 0$

**Answer (4)**

**Sol.** Given reaction, $2Cl(g) \longrightarrow Cl_2(g)$

We know that,

$Cl_2(g) \longrightarrow 2Cl(g)$ is endothermic reaction because it requires energy to break bond.

So reverse reaction is exothermic $\Delta_r H < 0$

Also, two gaseous atom combine together to form 1 gaseous molecule.

So, randomness $\Delta_r S < 0$

128. Measuring Zeta potential is useful in determining which property of colloidal solution?

(1) Viscosity

(2) Solubility

(3) Stability of the colloidal particles

(4) Size of the colloidal particles

**Answer (3)**

**Sol.** In colloidal solution, the potential difference between the fixed layer and the diffused layer of opposite charge is known as Zeta potential.

The presence of equal and similar charges on colloidal particles is largely responsible in providing stability to the colloidal solution.

129. Urea reacts with water to form A which will decompose to form B. B when passed through $Cu^{2+}$ (aq), deep blue colour solution C is formed. What is the formula of C from the following ?

(1) $CuSO_4$

(2) $[Cu(NH_3)_4]^{2+}$

(3) $Cu(OH)_2$

(4) $CuCO_3 \cdot Cu(OH)_2$

**Answer (2)**

**Sol.**

$$NH_2CONH_2 + H_2O \longrightarrow \underset{(A)}{(NH_4)_2CO_3} \xrightarrow{\Delta} \underset{(B)}{NH_3(g) + CO_2(g) + H_2O(l)}$$

$$\underset{(B)}{NH_3(g)} \xrightarrow{Cu^{2+}(aq)} \underset{\substack{(C)\\ \text{[Blue coloured solution]}}}{[Cu(NH_3)_4]^{2+}}$$

130. Match the following and identify the correct option.

| | |
|---|---|
| (a) $CO(g) + H_2(g)$ | (i) $Mg(HCO_3)_2 + Ca(HCO_3)_2$ |
| (b) Temporary hardness of water | (ii) An electron deficient hydride |
| (c) $B_2H_6$ | (iii) Synthesis gas |
| (d) $H_2O_2$ | (iv) Non-planar structure |

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| (1) | (iii) | (i) | (ii) | (iv) |
| (2) | (iii) | (ii) | (i) | (iv) |
| (3) | (iii) | (iv) | (ii) | (i) |
| (4) | (i) | (iii) | (ii) | (iv) |

**Answer (1)**

**Sol.**
- Mixture of CO and $H_2$ gases is known as water gas or synthesis gas.
- Temporary hardness of water is due to bicarbonates of calcium and magnesium.
- Diborane ($B_2H_6$) is an electron deficient hydride.
- $H_2O_2$ is non-planar molecule having open book like structure.

131. Match the following :

| Oxide | Nature |
|---|---|
| (a) CO | (i) Basic |
| (b) BaO | (ii) Neutral |
| (c) $Al_2O_3$ | (iii) Acidic |
| (d) $Cl_2O_7$ | (iv) Amphoteric |

is correct option?

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| (1) | (i) | (ii) | (iii) | (iv) |
| (2) | (ii) | (i) | (iv) | (iii) |
| (3) | (iii) | (iv) | (i) | (ii) |
| (4) | (iv) | (iii) | (ii) | (i) |

**Answer (2)**

**Sol.** CO : Neutral oxide

BaO : Basic oxide

$Al_2O_3$ : Amphoteric oxide

$Cl_2O_7$ : Acidic oxide

132. The rate constant for a first order reaction is $4.606 \times 10^{-3}\ s^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is :

(1) 100 s

(2) 200 s

(3) 500 s

(4) 1000 s

**Answer (3)**

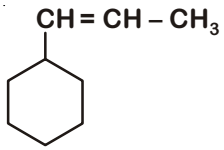
**Sol.** $k = \frac{2.303}{t}\log\frac{A_0}{A}$ (First order rate equation)

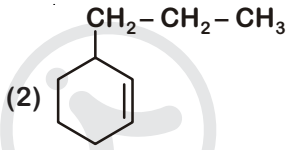
$$4.606\times10^{-3} = \frac{2.303}{t}\log\frac{2}{0.2}$$

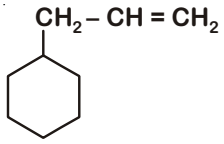
$$t = \frac{2.303}{4.606\times10^{-3}}\times\log10$$

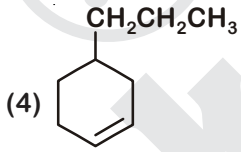
$$= \frac{10^3}{2} = 500 \text{ sec}$$

**133.** An alkene on ozonolysis gives methanal as one of the product. Its structure is

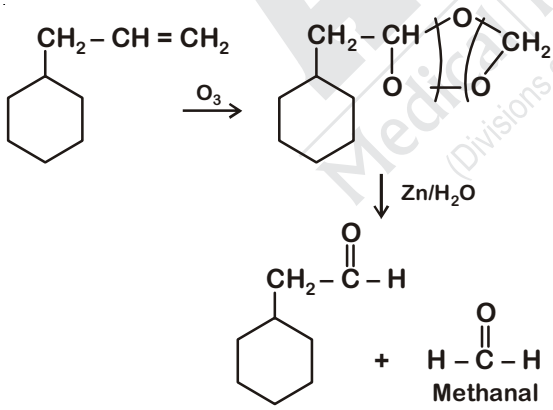
(1) 

(2) 

(3) 

(4) 

**Answer (3)**

**Sol.** 


**134.** Which of the following alkane cannot be made in good yield by Wurtz reaction?

(1) n-Hexane

(2) 2,3-Dimethylbutane

(3) n-Heptane

(4) n-Butane

**Answer (3)**

**Sol.** Wurtz reaction is used to prepare symmetrical alkanes like $R_1 - R_1$, as

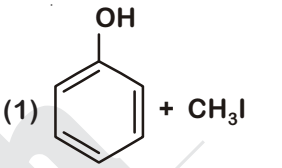
$$R_1 - X + 2Na + X - R_1 \xrightarrow{\text{Dry ether}} R_1 - R_1 + 2NaX$$

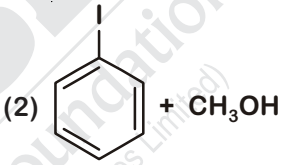
If $R_1$ and $R_2$ are different, then mixture of alkanes may be obtained as

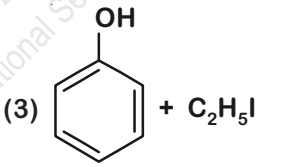
$$R_1 - X + 2Na + R_2 - X \xrightarrow{\text{Dry ether}}$$

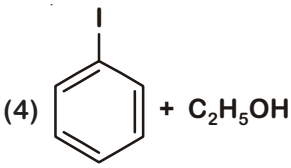
$$R_1 - R_1 + R_1 - R_2 + R_2 - R_2 + 2NaX$$

**135.** Anisole on cleavage with HI gives

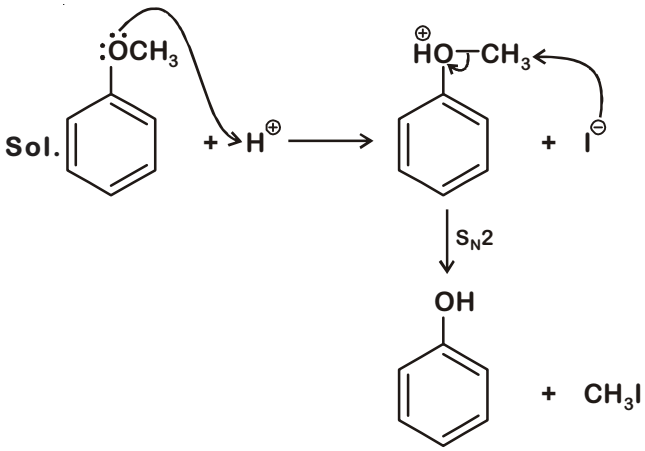
(1) 


(2) 


(3) 


(4) 


**Answer (1)**

**Sol.** 

136. For which one of the following, Bohr model is **not** valid ?

(1) Hydrogen atom

(2) Singly ionised helium atom ($He^+$)

(3) Deuteron atom

(4) Singly ionised neon atom ($Ne^+$)

**Answer (4)**

**Sol.** Bohr model is only valid for single electron species.

Singly ionised neon atom has more than one electron in orbit. Hence, Bohr model is not valid.

137. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is : (c = speed of electromagnetic waves)

(1) c : 1 (2) 1 : 1

(3) 1 : c (4) $1 : c^2$

**Answer (2)**

**Sol.** In an electromagnetic wave, half of the intensity is provided by the electric field and half by the magnetic field

Hence required ratio should be 1 : 1

138. The Brewsters angle $i_b$ for an interface should be

(1) $0° < i_b < 30°$ (2) $30° < i_b < 45°$

(3) $45° < i_b < 90°$ (4) $i_b = 90°$

**Answer (3)**

**Sol.** $\mu = \tan i_b$

$1 < \mu < \infty$

$1 < \tan i_b < \infty$

$\tan^{-1}(1) < i_b < \tan^{-1}(\infty)$

$45° < i_b < 90°$

139. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C.

Its density is : ($R = 8.3\ J\ mol^{-1}\ K^{-1}$)

(1) $0.5\ kg/m^3$ (2) $0.2\ kg/m^3$

(3) $0.1\ kg/m^3$ (4) $0.02\ kg/m^3$

**Answer (2)**

**Sol.** $PM = \rho RT \quad \Rightarrow \quad \rho = \frac{PM}{RT}$

$P = 249 \times 10^3\ N/m^2$

$M = 2 \times 10^{-3}\ kg$

$T = 300\ K$

$\therefore \quad \rho = \frac{(249\times10^3)(2\times10^{-3})}{8.3\times300} = \frac{0.2\ kg}{m^3}$

140. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is $\mu$, then the angle of incidence is nearly equal to :

(1) $\frac{A}{2\mu}$ (2) $\frac{2A}{\mu}$

(3) $\mu A$ (4) $\frac{\mu A}{2}$

**Answer (3)**

**Sol.** Light ray emerges normally from another surface, hence e(angle of emergence) = 0

$r_2 = 0$

$r_1 + r_2 = A$

$\Rightarrow \quad r_1 = A$

Applying Snell's law on first surface

$1.\sin i = \mu \sin r_1$

$\Rightarrow \quad \sin i = \mu \sin A$

For small angles ($\sin\theta \approx \theta$)

hence $i = \mu A$

141. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is :

(1) isothermal

(2) adiabatic

(3) isochoric

(4) isobaric

**Answer (2)**

**Sol.** Entire system is thermally insulated. So, no heat exchange will take place. Hence, process will be adiabatic.

142. The energy equivalent of 0.5 g of a substance is :

(1) $4.5 \times 10^{16}$ J

(2) $4.5 \times 10^{13}$ J

(3) $1.5 \times 10^{13}$ J

(4) $0.5 \times 10^{13}$ J

**Answer (2)**

**Sol.** From mass-energy equivalence.

$E = mc^2$

$= 0.5 \times 10^{-3} \times (3 \times 10^8)^2$

$= 4.5 \times 10^{13}$ J